BIRLA INSTITUTE OF TECHNOLOGY & SCIENCE, PILANI

WORK INTEGRATED LEARNING PROGRAMMES (WILP) DIVISION

DISSERTATION / PROJECT OUTLINE

FinAI – Advanced AI Based Personal Finance & Investment Analysis Tool

Course No.	**CSIWZG628T**
Course Title	**Dissertation / Project / Project Work**
Student Name	**Sugamanchi Naveen**
BITS ID	**2022WB86112**
Degree Program	**M.Tech in Computing Systems & Infrastructure (WIMS)**
Research Area	**Web Technologies**
Project carried out at	Wipro Limited, Bangalore
Supervisor	Soham Das
Additional Examiner	Smrutirekha Behera
Submission Month and Year	Oct 2026

Project Outline

Student ID No.	2022WB86112
Name of Student	Sugamanchi Naveen
Student Email Address	2022wb86112@wilp.bits-pilani.ac.in
Student's Employing Organization & Location	Wipro Limited, Bangalore
Supervisor's Name	Soham Das
Supervisor's Organization & Location	Wipro Technologies Ltd., Plot No. 1, 7, 8 & 9, Block DM, Sector V, Salt Lake City (Bidhannagar), Kolkata, West Bengal 700091
Supervisor's Email Address	soham.das@wipro.com
Additional Examiner's Name	Smrutirekha Behera
Additional Examiner's Organization & Location	Wipro Limited, Survey No. 124 & Part of 132/P, SEZ Gopanapally, Osman Nagar, Gopanapally, Hyderabad, Telangana 501301
Additional Examiner's Email Address	smrutirekha.behera@wipro.com

Project Title	**Advanced AI Based Personal Finance & Investment Analysis Tool**

1. Broad Academic Area of Work

- Artificial Intelligence and Machine Learning for financial analytics, anomaly detection, and sentiment analysis.
- Cloud-based full-stack web application development using Python FastAPI, PostgreSQL, and React.js with TypeScript.
- Financial modeling, investment valuation (DCF, Graham PEG, P/E multiples), and technical indicator analysis.
- Personal financial planning, budget management, savings tracking, and compound interest calculations (SIP, FD, XIRR).
- Data security, role-based access control, and compliant user data management.

2. Background

- Individuals often rely on fragmented tools such as basic spreadsheets and disconnected mobile apps to track expenses, monitor budgets, and evaluate investments.
- Traditional budgeting applications only record historical transactions without providing explainable insights or highlighting spending anomalies.
- Investment analysis tools are typically isolated from personal financial data, making it difficult for users to evaluate investment opportunities against their actual savings capacity.
- Many modern AI-driven financial platforms produce opaque recommendations without transparent mathematical calculations or verifiable data references.
- FinAI is proposed as a unified, cloud-based platform that combines intelligent personal financial management with transparent investment valuation and explainable AI insights.

3. Objectives

- Develop a centralized platform to record, categorize, and manage income, expenses, and recurring financial transactions.
- Implement configurable monthly budgets with real-time tracking and threshold alert notifications.
- Apply statistical anomaly detection algorithms to identify unusual spending spikes and irregular transactions.
- Formulate an explainable financial health scoring model based on savings rate, debt ratio, and budget adherence.
- Build quantitative equity valuation modules supporting Discounted Cash Flow (DCF), PEG, and market multiple models.
- Provide interactive financial planning calculators for Systematic Investment Plans (SIP), Fixed Deposits (FD), and wealth projections.
- Deploy a secure, cloud-hosted web application with role-based access control, audit logging, and responsive user dashboards.

4. Scope of Work

- User Management & Security: Registration, secure login, role-based access control, session management, and disclaimer tracking.
- Expense & Budget Module: Multi-category expense tracking, recurring transactions, income logging, and dynamic budget utilization progress.
- Analytics & AI Module: Monthly spending trends, category distributions, statistical Z-score anomaly detection, and automated financial insights.
- Investment & Valuation Module: Equity metrics, technical indicators (SMA, EMA, RSI, MACD), DCF valuation, and stock score generation.
- Financial Planning Module: SIP with step-up projections, Fixed Deposit compound growth, inflation adjustments, and CAGR/XIRR calculators.
- Technology Stack: Python FastAPI backend, PostgreSQL/SQLite database, React.js frontend with Tailwind CSS, and Docker deployment.

5. Plan of Work

The following is a proposed semester-level plan aligned with institute academic milestones and supervisor reviews:

Phase	Planned Work	Deliverable
Phase 1 – Requirements & Architecture	Finalize requirements, database schema, and system architecture. Implement user authentication, core income and expense tracking, and monthly budget management.	Approved project outline, system design specification, and core tracking application.
Phase 2 – Analytics & Anomaly Detection	Develop category-wise trend analytics, visual charts, statistical Z-score expense anomaly detection, and explainable financial health scoring.	Functional analytics dashboard with anomaly alerts and health score module.
Phase 3 – Investment & Valuation Engines	Implement financial planning calculators (SIP, FD, CAGR, XIRR) and equity valuation models (DCF, Graham PEG, P/E multiples, and technical indicators).	Interactive investment analysis tools, valuation engine, and stock analysis reports.
Phase 4 – AI Assistant & System Integration	Integrate grounded AI assistant with data citations for market queries, complete end-to-end integration, performance optimization, and containerized deployment.	Integrated full-stack AI financial platform ready for cloud deployment.
Phase 5 – Testing, Evaluation & Dissertation	Perform functional testing, user evaluation, and accuracy validation of financial models. Prepare final dissertation report, slides, and viva demonstration.	Completed dissertation thesis, presentation slides, and viva demonstration.

6. Literature References

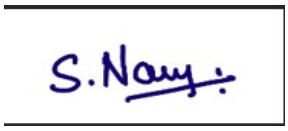
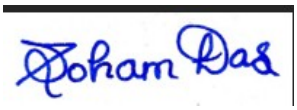
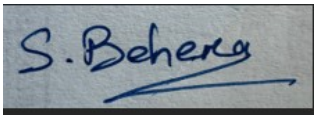
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7. Particulars of the Supervisor and Additional Examiner

Particulars	Supervisor	Additional Examiner
Name	Soham Das	Smrutirekha Behera
Qualification	Bachelor of Technology (B.Tech) in Computer Science and Engineering (CSE)	Master of Technology (M.Tech) in Software Engineering
Designation	Senior Technical Specialist (7 years of IT experience)	Technical Consultant (12 years of IT experience)
Employing organization and location	Wipro Technologies Ltd., Plot No. 1, 7, 8 & 9, Block DM, Sector V, Salt Lake City (Bidhannagar), Kolkata, West Bengal 700091	Wipro Limited, Survey No. 124 & Part of 132/P, SEZ Gopanapally, Osman Nagar, Gopanapally, Hyderabad, Telangana 501301
Phone number	+91 7073143358	+91 8885489246
Email address	soham.das@wipro.com	smrutirekha.behera@wipro.com

8. Remarks of the Supervisor

[For supervisor's comments and approval.]

Student Signature	Supervisor Signature	Additional Examiner Signature
		
Name: Sugamanchi Naveen Date: 27/10/2026	Name: Soham Das Date: 27/10/2026	Name: Smrutirekha Behera Date: 27/10/2026